Title Of The Project

SAFE LINKER

Project - I I

Group - F



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Agenda

- Introduction
- Problem Statement
- Objectives
- Flowchart
- Methodology
- Problem Faced
- Suggestion from Mentor
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Introduction

- SAFE LINKER is a real-time system that detects phishing and harmful links using statistical analysis and machine learning, classifying URLs as Safe or Unsafe based on key features.

- The system is to be developed using Python .it will be test Urls directly within a Jupyter Notebook that helps users identify risky links and ensures safer browsing.

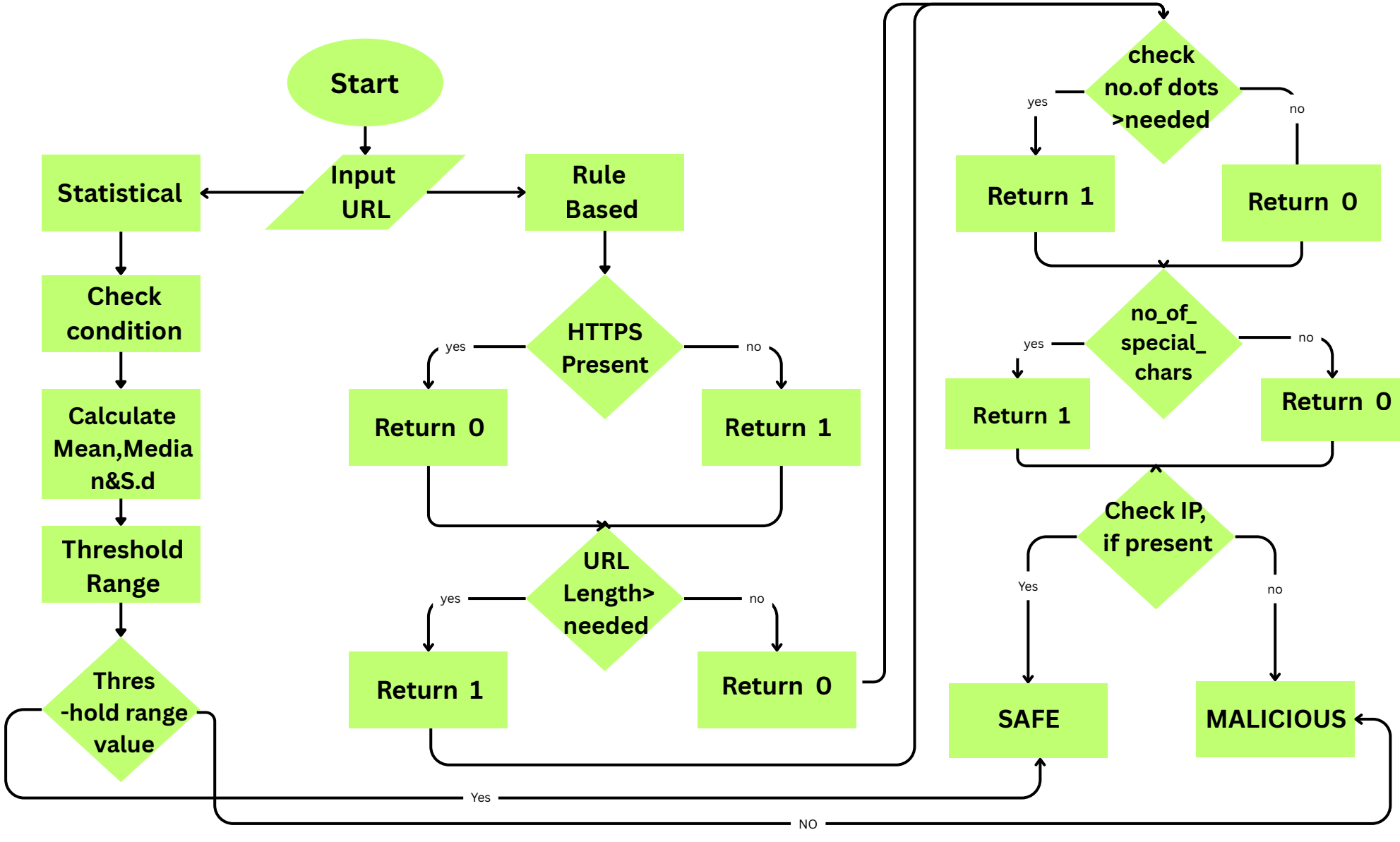
Problem Statement

To develop an ML-based model for detecting malicious URLs.

Objectives

- To collect and preprocess URL datasets.
- To identify statistical features that detect malicious URLS.
- To implement machine learning algorithms for rule-based and Statistical URL classification.
- To test the ML based model's performance and generate the final result

Flowchart


Start
Input URL
Statistical
Rule Based
Check condition
Calculate Mean,Median&S.d
Threshold Range
Thres -hold range value
HTTPS Present
yes
no
Return 0
Return 1
URL Length> needed
yes
no
Return 1
Return 0
check no.of dots >needed
yes
no
Return 1
Return 0
no_of_ special_ chars
yes
no
Return 1
Return 0
Check IP, if present
Yes
no
SAFE
MALICIOUS
Yes
NO

Methodology

1.Data Collection: Collect safe and malicious URL datasets from Kaggle.

2.Feature Extraction: Identify and extract features of each URL such as length, IP address, and number of dots etc.

3.Rule-Based & Statistical Classification: Identify URL patterns, apply rule-based and statistical thresholds, and use 3 algorithms (Random Forest Classifier, Logistic Regression Classifier,Naive Bayes Classifier) to build a machine learning model for classification.

4.Testing And Result: Check URLs in real-time and test it using both real and sample URLs.

Problem Faced

- **Dependency & Version Errors:** The model show error due to missing or mismatched libraries; the issue was resolved by installing and updating the required packages .

- **ValueError during Model Training :** Faced a ValueError because of mismatched data size and wrong data format during model training.

- **Path Confusion:** The file path configuration caused errors, which were resolved by correcting the file paths.

Suggestions from mentor:

- HTTPS links can be malicious.
- Google Drive links show unsafe cause of url length.
- Detection conditions are limited.
- Add and compare all 3 Algorithms.
- Take different size dataset and compare with each other.

Technology Stack









References

IEEE : Y. A. Kustiawan and K. I. Ghauth, "Evaluating the Impact of Feature Engineering in Phishing URL Detection: A Comparative Study of URL, HTML, and Derived Features," IEEE Access, vol. 13, pp. 126756–126768, Jan. 2025, doi: 10.1109/ACCESS.2025.3579223. [Online]. Available: https://ieeexplore.ieee.org/document/11031414

Kaggle : https://www.kaggle.com/datasets/shashwatwork/web-page-phishing-detection-dataset



Thank You